Project Report

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1 Introduction

We study the least-squares problem for a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $b \in \mathbb{R}^m$.

2 Method

The estimate minimizes the residual norm,

$$\hat{x} = \arg\min_{x \in \mathbb{R}^n} \|Ax - b\|^2 ,$$

and satisfies the normal equations $A^\top A\hat{x} = A^\top b$.